FURTHER MECHANICS / KINEMATICS & DYNAMICS

# Impulse and Work Done by a Variable Force

Mark scheme 11 questions · 123 marks

EXAM BOARDS AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421

## Questions


| Question | Marks by part | Marks | Page |
|---|---|---|---|
| Question 1 | (a) 4 (b) 3 | 7 | 4 |
| Question 2 | (a) 2 (b) 5 | 7 | 5 |
| Question 3 | (a) 3 (b) 3 | 6 | 6 |
| Question 4 | (a) 3 (b)(i) 3 (b)(ii) 2 | 8 | 7 |
| Question 5 | (a) 3 (b) 2 (c) 5 | 10 | 8 |
| Question 6 | (a) 3 (b) 3 (c) 4 | 10 | 9 |
| Question 7 | (a) 3 (b) 2 (c) 6 | 11 | 10 |
| Question 8 | (a) 5 (b) 5 (c) 2 (d) 1 | 13 | 11 |
| Question 9 | (a) 5 (b) 5 (c) 5 | 15 | 12 |
| Question 10 | (a) 8 (b) 8 | 16 | 14 |
| Question 11 | (a) 6 (b) 5 (c) 5 (d) 4 | 20 | 16 |
| | **Total** | **123** | |

# Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

### Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

| Code | Meaning |
|---|---|
| **M1** | Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted. |
| **dM1** | Dependent method: as M1, but only if the method mark before it was earned, unless its note names another. |
| **A1** | Accuracy: a correct answer or step. It needs the method mark it follows. |
| **A1ft** | As A1, following through from an earlier answer that the note names. |
| **A1 AG** | The last mark of a part whose answer is printed in the question: complete working, with no errors. |
| **B1** | A correct value, fact or statement. It needs no method mark. |
| **B1ft** | As B1, following through from an earlier answer that the note names. |
| **E1** | A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark. |

### Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

| | | | |
|---|---|---|---|
| **ft** | follow through<br>Correct work from the student's own earlier wrong value earns the mark. | **AG** | answer given<br>The answer is printed in the question, so every step to it must be shown. |
| **cao** | correct answer only<br>No follow-through from an earlier error. | **cso** | correct solution only<br>No error anywhere in the part, not just in its answer. |
| **oe** | or equivalent | **awrt** | answers which round to |
| **soi** | seen or implied<br>Earned if written, or if later working shows it was used. | **isw** | ignore subsequent working<br>A correct answer, once reached, isn't spoiled by later working. |
| **SC** | special case<br>Credit, as the note says, for work outside the main method. | **their** | the student's own earlier value |
| **condone** | a slip that doesn't cost the mark | [ ] | working the mark doesn't need |
| **s.f.** | significant figures | **d.p.** | decimal places |

### Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

### Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

### Value of $g$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state $g$, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **1(a)** | $\mathbf{I} = 0.40(12\mathbf{j}) - 0.40(9\mathbf{i})$ | **M1** | Uses $m(\mathbf{v} - \mathbf{u})$ with both velocities as vectors; condone the subtraction reversed. Not for subtracting speeds. |
| | $= (-3.6\mathbf{i} + 4.8\mathbf{j})\,\mathrm{N\,s}$ | **A1** | oe, such as a column vector. |
| | $\lvert\mathbf{I}\rvert = \sqrt{(-3.6)^2 + 4.8^2}$ | **M1** | Uses Pythagoras on both components of their impulse vector. |
| | $= 6\,\mathrm{N\,s}$ | **A1** | |
| | | **[4]** | |
| **1(b)** | $\int_0^{0.040} k\sin(25\pi t)\,\mathrm{d}t =$ "6"<br>$\Rightarrow \left[-\frac{k}{25\pi}\cos(25\pi t)\right]_0^{0.040} =$ "6" | **M1** | Equates the integral of $F$ over $0 \leq t \leq 0.040$ to their magnitude from (a) and integrates to the form $p\cos(25\pi t)$. Condone sign and coefficient slips. |
| | $\frac{k}{25\pi}(\cos 0 - \cos \pi) = \frac{2k}{25\pi}$ | **A1** | Correct integral with both limits applied, oe. |
| | $k = 75\pi$ | **A1** | cao |
| | | **[3]** | |
| **Total for Question 1** | | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **2(a)** | $I = 6(0) - 6(4)\ [= -24]$ | **M1** | Uses impulse as change in momentum with mass 6 and speeds 4 and 0. Condone a sign slip. |
| | $24\,\text{N s}$ | **A1** | Must be positive. |
| | | **[2]** | |
| **2(b)** | $\int_0^T 18\,000t(T-t)\,\mathrm{d}t =$ "24"<br>$\Rightarrow 18\,000\left[\frac{Tt^2}{2} - \frac{t^3}{3}\right]_0^T =$ "24" | **M1** | Equates the integral of $F$ from 0 to $T$ to their impulse from (a) and integrates, at least one power increased by 1. Condone sign slips. |
| | $3000T^3 = 24$ | **A1** | Correct integral, $3000T^3$ oe. |
| | $T = 0.2$ | **A1** | |
| | $\frac{\mathrm{d}F}{\mathrm{d}t} = 18\,000(T-2t) = 0 \Rightarrow t = \ldots$ | **M1** | Differentiates $F$, at least one power reduced by 1, and sets it to zero, or uses the quadratic's symmetry, to find $t$ in terms of $T$ or their $T$. |
| | $F_{\max} = 18\,000(0.1)(0.2-0.1) = 180\,\text{N}$ | **A1** | |
| | | **[5]** | |
| | **Total for Question 2** | **7** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **3(a)** | $W = \int_0^{0.5} \left(\frac{A}{1+2x} + 200x\right) \mathrm{d}x = \ldots$ | **M1** | Integrates $F$ with respect to $x$ between 0 and 0.5, at least one term integrated correctly. Limits may be applied later. |
| | $\left[\frac{A}{2}\ln(1+2x) + 100x^2\right]_0^{0.5}$ | **A1** | Both terms correct; logarithm term oe, such as $\frac{A}{2}\ln\left(x+\frac{1}{2}\right)$. |
| | $W = \frac{A}{2}\ln 2 + 25$ (J) | **A1** | oe, such as $A\ln\sqrt{2}+25$, or awrt $0.347A + 25$. |
| | | **[3]** | |
| **3(b)** | $\frac{1}{2}(20)(3)^2 + (20)(9.8)(0.5\sin 30°)$ $-$ "$\frac{A}{2}\ln 2 + 25$" $= 0$ | **M1 A1** | **M1** Forms an energy equation with initial KE, GPE using a resolved drop, and their (a) of the form $pA + q$. Condone sign slips and sin/cos confusion. **A1** Correct equation, with (a) correct. |
| | $A = \frac{228}{\ln 2}$ | **A1** | Any exact form, such as $\frac{456}{\ln 4}$. |
| | | **[3]** | |
| | **Total for Question 3** | **6** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **4(a)** | $I - 40 \times 2 = 80(7 - 3)$ | **M1 A1** | **M1** Forms an impulse–momentum equation with the launcher's impulse, the resistance's impulse $40 \times 2$ and the change in momentum. Condone sign slips.<br>**A1** Correct equation, oe. |
| | $I = 400\,\text{N s}$ | **A1** | |
| | | **[3]** | |
| **4(b)(i)** | "400" $= \int_0^2 kt^2(2 - t)\,\mathrm{d}t$<br>$= k\left[\frac{2t^3}{3} - \frac{t^4}{4}\right]_0^2$ | **M1** | Equates $\int_0^2 F\,\mathrm{d}t$ to their impulse from (a) and integrates, the power of at least one term increased by 1. |
| | $= \frac{4k}{3}$ | **A1** | Correct integral with limits applied, oe. |
| | $k = 300$ | **A1** | |
| | | **[3]** | |
| **4(b)(ii)** | $\frac{\mathrm{d}F}{\mathrm{d}t} =$ "300"$(4t - 3t^2) = 0 \Rightarrow t = \ldots$ $[= \frac{4}{3}]$<br>$F =$ "300" $\left(\frac{4}{3}\right)^2\left(2 - \frac{4}{3}\right)$ | **M1** | Differentiates their $F$, the power of at least one term decreased by 1, sets it to zero and substitutes a nonzero root into $F$. |
| | $F_{\max} = \frac{3200}{9}\,\text{N}$ | **A1** | $\frac{3200}{9}$ or awrt 356. |
| | | **[2]** | |
| **Total for Question 4** | | **8** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **5(a)** | $I = \int_0^2 (3t^2 - 12t + 9)\,\mathrm{d}t$ | **M1** | Integrates $F$ with respect to $t$ between 0 and 2, with at least one power increased by 1. |
| | $= \left[t^3 - 6t^2 + 9t\right]_0^2$ | **A1** | |
| | $I = 2\,\mathrm{N\,s}$ | **A1** | |
| | | **[3]** | |
| **5(b)** | "2" $= 2v - 2(3) \Rightarrow v = \ldots$ | **M1** | Uses impulse = change in momentum with their impulse from (a), mass 2 and initial velocity 3. |
| | $4\,\mathrm{m\,s^{-1}}$ | **A1ft** | ft $3 + \frac{1}{2}$ their impulse from (a). |
| | | **[2]** | |
| **5(c)** | $3t^2 - 12t + 9 = 0 \Rightarrow t = 1, 3$<br>Force opposes the motion for $1 < t \leq 2$ | **B1** | Interval from $t = 1$ to $t = 2$ identified; strictness of inequalities condoned. |
| | $\int_0^1 (3t^2 - 12t + 9)\,\mathrm{d}t = 2v_1 - 2(3) \Rightarrow v_1 = \ldots$ | **M1** | Equates an integral of $F$ over 0 to 1 (or 1 to 2, with their speed at $t = 2$) to the change in momentum. Condone slips. |
| | $v_1 = 5\,\mathrm{m\,s^{-1}}$ | **A1** | |
| | $\frac{1}{2}(2)\,(\text{"5"})^2 + W = \frac{1}{2}(2)\,(\text{"4"})^2$ | **M1** | Equates the work done to the change in KE between $t = 1$ and $t = 2$, with their speeds at both times. Condone sign slips. |
| | $W = -9\,\mathrm{J}$ | **A1** | $-9\,\mathrm{J}$, or $9\,\mathrm{J}$ stated as work done against the motion. |
| | | **[5]** | |
| | **Total for Question 5** | **10** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **6(a)** | $I = \int_0^2 (3t^2 - 12t)\,\mathrm{d}t$<br>$= \left[t^3 - 6t^2\right]_0^2$ | **M1 A1** | **M1** Integrates $F$ with respect to $t$, at least one power increased by 1, and uses the limits 0 and 2.<br>**A1** Correct integral, oe. |
| | $I = -16 \Rightarrow$ magnitude $16\,\mathrm{N\,s}$ | **A1** | A0 if the magnitude is left negative. |
| | | **[3]** | |
| **6(b)** | $4\frac{\mathrm{d}v}{\mathrm{d}t} = 3t^2 - 12t \Rightarrow 4v = t^3 - 6t^2 + c$ | **M1** | Uses $F = ma$ with $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, or impulse equals change in momentum, and integrates $F$ with respect to $t$. Condone a missing constant. |
| | $v = 4,\ t = 0 \Rightarrow 4v = t^3 - 6t^2 + 16$ | **A1** | oe, such as $v = \frac{1}{4}t^3 - \frac{3}{2}t^2 + 4$, or $-27 = 4v - 16$ from the impulse over $0 \leq t \leq 3$. |
| | $t = 3 \Rightarrow v = -\frac{11}{4}$<br>Speed $\frac{11}{4}\,\mathrm{m\,s^{-1}}$, negative direction | **A1** | Speed $\frac{11}{4}$ oe, given as positive, and the negative direction stated. |
| | | **[3]** | |
| **6(c)** | "$t^3 - 6t^2 + 16$" $= 0$ | **M1** | Sets their $v$ from (b), a cubic in $t$, equal to zero. |
| | $(t-2)(t^2 - 4t - 8) = 0$ | **dM1 A1** | **dM1** Takes out the factor $(t-2)$ by division, inspection or comparing coefficients, reaching a quadratic factor.<br>**A1** Correct quadratic factor. |
| | $t^2 - 4t - 8 = 0 \Rightarrow t = 2 \pm 2\sqrt{3}$<br>$t = 2 + 2\sqrt{3}\,\mathrm{s}$ | **A1** | Exact, oe such as $2 + \sqrt{12}$. A0 if $2 - 2\sqrt{3}$ is also given as an answer. |
| | | **[4]** | |
| | **Total for Question 6** | **10** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **7(a)** | $I = \int_0^1 (6t^2 - 24t + 18)\,\mathrm{d}t$<br>$= \left[2t^3 - 12t^2 + 18t\right]_0^1$ | **M1 A1** | **M1** Integrates $F$ with respect to $t$, the power of at least one term increased by 1, with limits 0 and 1 used.<br>**A1** Correct integral, oe. |
| | $8\,\mathrm{N\,s}$ | **A1** | |
| | | **[3]** | |
| **7(b)** | "8" $= 4v - 4(-1) \Rightarrow v = \ldots$ | **M1** | Uses impulse equals change in momentum with $m = 4$, $u = -1$ and their impulse from (a). |
| | $v = 1\,\mathrm{m\,s^{-1}}$ | **A1** | |
| | | **[2]** | |
| **7(c)** | $4\frac{\mathrm{d}v}{\mathrm{d}t} = 6(t-1)(t-3) \Rightarrow v = \ldots$ | **M1** | Divides by the mass and integrates with respect to $t$, the power of at least one term increased by 1. |
| | $v = \frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t + c$ | **A1** | |
| | "1" $= \frac{1}{2} - 3 + \frac{9}{2} + c \Rightarrow c = \ldots$ | **dM1** | Uses their velocity at $t = 1$ from (b) to find $c$; a definite integral from $t = 1$ is equivalent. |
| | $v = \frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t - 1$ | **A1** | oe |
| | $v = 1$: $t^3 - 6t^2 + 9t - 4 = 0 \Rightarrow t = \ldots$<br>$v = -1$: $t^3 - 6t^2 + 9t = 0 \Rightarrow t = \ldots$ | **M1** | Sets their cubic $v$ equal to 1 and to $-1$ and solves both, by any means. |
| | $t = 1,\ 3,\ 4$ | **A1** | These three only; A0 if $t = 0$ is given as an answer. |
| | | **[6]** | |
| | **Total for Question 7** | **11** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **8(a)** | $W = \int_0^h \left(498 + \frac{k}{(x+2)^2}\right) \mathrm{d}x$<br>$= \left[498x - \frac{k}{x+2}\right]_0^h$ | **M1 A1** | **M1** Integrates $P$ with respect to $x$, at least one term correct, and uses limits 0 and $h$.<br>**A1** Correct integral, oe; limits need not be substituted yet. |
| | $= 498h + \frac{k}{2} - \frac{k}{h+2}$ | **A1** | oe, such as $498h + \frac{kh}{2(h+2)}$. |
| | $498 + \frac{k}{(x+2)^2} = 60(9.8) \Rightarrow x = \ldots$ | **M1** | Sets $P$ equal to the weight 588 and rearranges for $x$. Condone slips in the rearrangement. |
| | $x = \sqrt{\frac{k}{90}} - 2$ | **A1** | oe. Positive root only. |
| | | **[5]** | |
| **8(b)** | "$498x + k\left(\frac{1}{2} - \frac{1}{x+2}\right)$"<br>$= \frac{1}{2}(60)v^2 + 588x$ | **M1** | Forms an energy equation, at general $x$ or at $x = 6$, with their work from (a) in terms of $k$ and $x$, KE and the GPE gained. Condone sign slips. |
| | $30(3)^2 = -90(6) + k\left(\frac{1}{2} - \frac{1}{8}\right)$ | **A1** | oe. |
| | $k = 2160$ | **A1** | |
| | $30v^2 = -90(8) +$ "2160" $\left(\frac{1}{2} - \frac{1}{10}\right)$<br>$\Rightarrow v = \ldots$ | **dM1** | Substitutes $x = 8$ and their $k$ into their energy equation and solves for $v$. |
| | $v = 2.19\,\mathrm{m\,s^{-1}}$ | **A1** | awrt 2.19. |
| | | **[5]** | |
| **8(c)** | $0 = -90h +$ "2160" $\left(\frac{1}{2} - \frac{1}{h+2}\right)$<br>$\Rightarrow h = \ldots$ | **M1** | Sets $v = 0$ in their energy equation from (b), with their $k$, and solves for a nonzero $h$. |
| | $h = 10\,\mathrm{m}$ | **A1** | |
| | | **[2]** | |
| **8(d)** | Small frontal area relative to its mass, so air resistance is small compared with its weight | **E1** | Gives a reason why air resistance is small beside the weight, such as small frontal area relative to mass, high density or low speed. Not for "compact" alone. |
| | | **[1]** | |
| | **Total for Question 8** | **13** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **9(a)** | $I = k\int_0^{0.1}(0.2t - t^2)\,\mathrm{d}t$<br>$\quad + k\int_{0.1}^{0.2}(0.2 - t)^2\,\mathrm{d}t$<br>$= k\left[0.1t^2 - \frac{t^3}{3}\right]_0^{0.1}$<br>$\quad + k\left[0.04t - 0.2t^2 + \frac{t^3}{3}\right]_{0.1}^{0.2}$ | **M1 A1** | **M1** Splits the integral at $t = 0.1$ and integrates both pieces, at least one term of each increased in power. Condone a missing $k$.<br>**A1** Both correct, oe such as $-\frac{(0.2-t)^3}{3}$. |
| | $= k\left(0.001 - \frac{0.001}{3}\right) + k\left(\frac{0.008}{3} - \frac{0.007}{3}\right)$ | **dM1** | Substitutes both limits into each of their integrated pieces and subtracts. |
| | $= \frac{k}{1500} + \frac{k}{3000}$ | **A1** | |
| | $= \frac{k}{1000}\,\mathrm{N\,s}$ | **A1 AG** | Correct integrals and limits leading without error to $\frac{k}{1000}$. Numerical integration alone earns nothing. |
| | | **[5]** | |
| **9(b)** | $u = 5\sin 60° \quad \left[= \frac{5\sqrt{3}}{2}\right]$ | **B1** | Perpendicular approach component, oe such as $5\cos 30°$, either sign. |
| | $\frac{k}{1000} = 0.4\left(e \times \text{“}\frac{5\sqrt{3}}{2}\text{”}\right)$<br>$\quad + 0.4\left(\text{“}\frac{5\sqrt{3}}{2}\text{”}\right)$ | **M1** | Equates the impulse from (a) to the change in perpendicular momentum, with rebound component $e$ times their approach component. Condone sign slips. |
| | $k = 1000\sqrt{3}(e + 1)$ | **A1** | oe. |
| | $k$ increases with $e$, so<br>$1000\sqrt{3}\left(1 + \frac{1}{4}\right) \le k \le 1000\sqrt{3}\left(1 + \frac{3}{4}\right)$ | **dM1** | Substitutes $e = \frac{1}{4}$ and $e = \frac{3}{4}$ into their linear $k$ in $e$, or carries the bounds on $e$ through to $k$, or makes $e$ the subject and applies its bounds. |
| | $1250\sqrt{3} \le k \le 1750\sqrt{3}$ | **A1** | Exact bounds oe, both inequalities non-strict, justified by $k$ increasing with $e$ (a positive coefficient) or by the bounds carried through. |
| | | **[5]** | |
| **9(c)** | $1500\sqrt{3} = \text{“}1000\sqrt{3}(e + 1)\text{”}$<br>$\Rightarrow e = \ldots \quad \left[= \frac{1}{2}\right]$<br>$v = e \times \text{“}\frac{5\sqrt{3}}{2}\text{”}$ | **M1** | Finds the perpendicular rebound component, from $e$ by their $k$ in terms of $e$ from (b), or directly from impulse $1.5\sqrt{3}$ equal to the change in momentum. |
| | $v = \frac{5\sqrt{3}}{4}$ | **A1** | oe, or awrt 2.17. |

*Question 9 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $V = \sqrt{\left(\text{“}\frac{5\sqrt{3}}{4}\text{”}\right)^2 + (5\cos 60°)^2}$<br>$\tan\theta = \text{“}\frac{5\sqrt{3}}{4}\text{”} \div 5\cos 60°$ | **M1** | Uses Pythagoras for the speed, or a trigonometric ratio for the angle, with their perpendicular component and the parallel component unchanged by the impact. Condone sin/cos confusion. |
| | $V = \frac{5\sqrt{7}}{4}\,\text{m}\,\text{s}^{-1}$ | **A1** | Exact oe, such as $\frac{\sqrt{175}}{4}$. |
| | $\theta = 40.9°$ | **A1** | awrt 40.9°, measured from the wall. |
| | | **[5]** | |
| | **Total for Question 9** | **15** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **10(a)** | $800v\frac{\mathrm{d}v}{\mathrm{d}x} = 1800 + 1200e^{-x/200} - 2v^2$ | **M1 A1** | **M1** Uses Newton's second law with $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, the driving force and the resistance. Condone sign slips.<br>**A1** Correct equation, oe. |
| | $\frac{\mathrm{d}u}{\mathrm{d}x} = 2v\frac{\mathrm{d}v}{\mathrm{d}x}$<br>$\Rightarrow 400\frac{\mathrm{d}u}{\mathrm{d}x} = 1800 + 1200e^{-x/200} - 2u$ | **M1 A1** | **M1** Replaces $v\frac{\mathrm{d}v}{\mathrm{d}x}$ by a multiple of $\frac{\mathrm{d}u}{\mathrm{d}x}$ and $v^2$ by $u$, reaching an equation in $u$ and $x$ only. Condone a wrong multiple.<br>**A1** Correct equation in $u$ and $x$, oe. |
| | $\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{u}{200} = \frac{9}{2} + 3e^{-x/200}$<br>$\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(ue^{x/200}\right) = \frac{9}{2}e^{x/200} + 3$<br>$\Rightarrow ue^{x/200} = \ldots$ | **M1** | Multiplies their linear equation in standard form by its integrating factor and integrates, at least one RHS term correct. Condone coefficient slips. |
| | $ue^{x/200} = 900e^{x/200} + 3x + c$ | **A1** | Correct, including the constant, oe. |
| | $x = 0,\ u = 400 \Rightarrow 400 = 900 + c \Rightarrow c = \ldots$ | **dM1** | Substitutes $x = 0$, $u = 400$ (or $v = 20$ in $v^2$) into their solution with a constant to find it. Not for $u = 20$. |
| | $c = -500 \Rightarrow v^2 = 900 + (3x - 500)e^{-x/200}$ | **A1 AG** | Correct $c$ in a correct general solution, reaching the printed result in $v^2$ with no errors in the part. |
| | | **[8]** | |
| **10(b)** | $\int_0^{200}\left(1800 + 1200e^{-x/200}\right)\mathrm{d}x$<br>$= \left[1800x - 240\,000e^{-x/200}\right]_0^{200}$ | **M1 A1** | **M1** Integrates the driving force with respect to $x$ between 0 and 200, at least one term integrated correctly.<br>**A1** Correct integral; limits need not be substituted yet. |
| | $= 600\,000 - \frac{240\,000}{e}$ | **A1** | Exact oe, or awrt 512 000. |
| | $x = 200 \Rightarrow v^2 = 900 + \frac{100}{e}$ | **B1** | Exact oe, or awrt 937. |
| | $\frac{1}{2}(800)(20^2) + \text{“}600\,000 - \frac{240\,000}{e}\text{”} - W$<br>$= \frac{1}{2}(800)\text{“}\left(900 + \frac{100}{e}\right)\text{”}$ | **M1 A1** | **M1** Forms an energy equation with both KE terms, their work by the driving force and the work against resistance. Condone sign slips.<br>**A1** Correct equation, oe. |
| | $W = 400\,000 - \frac{280\,000}{e}$ | **A1** | Any exact equivalent. |

*Question 10 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $\frac{\text{“}W\text{”}}{200} = 2000 - \frac{1400}{e}$ | **B1ft** | ft their work against resistance divided by 200. Exact, or awrt 1480; also accept 1485 (4 s.f.). |
| | | **[8]** | |
| | **Total for Question 10** | **16** | |

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| **11(a)** | $2v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{27}{v} - 3v$ | **M1** | Uses Newton's second law with acceleration $v\frac{\mathrm{d}v}{\mathrm{d}x}$, mass 2 and the given force. Condone sign slips. |
| | $\frac{\mathrm{d}x}{\mathrm{d}v} = \frac{2v^2}{3(9-v^2)}$<br>$= -\frac{2}{3} + \frac{1}{3-v} + \frac{1}{3+v}$ | **A1** | Correct, oe, in any form such as $\frac{2v^2}{3(9-v^2)}$; partial fractions are not needed for this mark. |
| | $x = \int \left(-\frac{2}{3} + \frac{1}{3-v} + \frac{1}{3+v}\right) \mathrm{d}v$<br>$= -\frac{2}{3}v - \ln(3-v) + \ln(3+v) + c$ | **M1 A1** | **M1** Separates variables and integrates their expression in $v$, at least one term integrated correctly. Condone sign slips in the logarithms.<br>**A1** Correct, oe; $+c$ may be omitted here. |
| | $0 = -\frac{2}{3} - \ln 2 + \ln 4 + c \Rightarrow c = \ldots$ | **dM1** | Substitutes $x = 0$, $v = 1$ into their integrated equation to find the constant, or uses limits 0 to $x$ and 1 to $v$. |
| | $c = \frac{2}{3} - \ln 2$<br>$\Rightarrow x = \ln\left(\frac{3+v}{2(3-v)}\right) - \frac{2}{3}(v-1)$ | **A1 AG** | Correct constant substituted, and log laws leading without error to the printed result; combined steps accepted. The range need not be justified. |
| | | **[6]** | |
| **11(b)** | $x = \ln\frac{5}{2} - \frac{2}{3}$ | **B1** | oe, exact. |
| | $W = \int_1^2 \left(\frac{27}{v} - 3v\right) \frac{2v^2}{3(9-v^2)} \mathrm{d}v$<br>$= \int_1^2 2v \,\mathrm{d}v = \ldots$ | **M1** | Writes $\int F \,\mathrm{d}x$ as $\int F \frac{\mathrm{d}x}{\mathrm{d}v} \mathrm{d}v$ with their $\frac{\mathrm{d}x}{\mathrm{d}v}$ from (a), or as $\int 2v \,\mathrm{d}v$, and integrates at least one term correctly. |
| | $W = 3\,\mathrm{J}$ | **A1** | |
| | $F_{\text{av}} = \frac{\text{“3”}}{\text{“}\ln\frac{5}{2} - \frac{2}{3}\text{”}}$ | **M1** | Divides their work by their displacement at $v = 2$, exact or decimal. |
| | $F_{\text{av}} = 12.0\,\mathrm{N}$ | **A1** | awrt 12.0. |
| | | **[5]** | |
| **11(c)** | $2\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{27}{v} - 3v$ | **M1** | Uses Newton's second law with acceleration $\frac{\mathrm{d}v}{\mathrm{d}t}$, mass 2 and the given force. Condone sign slips. |
| | $\int \frac{2v}{9-v^2} \mathrm{d}v = \int 3 \,\mathrm{d}t$<br>$\Rightarrow -\ln(9-v^2) = 3t + k$ | **M1 A1** | **M1** Separates variables and integrates the $v$ side to a logarithm of $9 - v^2$, $27 - 3v^2$ or their factors. Condone a wrong coefficient.<br>**A1** Correct, oe; $+k$ may be omitted here. |

*Question 11 continued*

| Question | Scheme | Marks | Notes |
|---|---|---|---|
| | $-\ln(9-1) = 0 + k \Rightarrow k = \ldots$ | **dM1** | Substitutes $v = 1$, $t = 0$ into their integrated equation to find the constant, or uses limits 1 to $v$ and 0 to $t$. |
| | $k = -\ln 8 \Rightarrow v = \sqrt{9 - 8e^{-3t}}$ | **A1** | oe, with $v$ the subject and the positive root only. |
| | | **[5]** | |
| **11(d)** | $v(1) = \sqrt{9 - 8e^{-3}} = 2.933$ | **B1ft** | ft their $v$ from (c) at $t = 1$, to 4 s.f.; awrt 2.933 for the correct $v$. |
| | $v(2) = \sqrt{9 - 8e^{-6}} = 2.997$ | **B1ft** | ft their $v$ from (c) at $t = 2$, to 4 s.f.; awrt 2.997 for the correct $v$. |
| | $I = 2\,(\text{“}2.997\text{”} - \text{“}2.933\text{”})$ | **M1** | Multiplies 2 by the difference of their two velocities. Condone subtraction in the wrong order. |
| | $I = 0.128\,\text{N s}$ | **A1** | awrt 0.128, positive. |
| | | **[4]** | |
| | **Total for Question 11** | **20** | |